```
import numpy as np
#creating 5*4 array

array = np.arange(20).reshape(5, 4)
print(array)
print()
```

```
[[ 0  1  2  3]
 [ 4  5  6  7]
 [ 8  9 10 11]
 [12 13 14 15]
 [16 17 18 19]]
```

```
print(np.argmax(array))
```

```
19
```

```
print(np.argmax(array, axis=1))
```

```
[3 3 3 3 3]
```

```
print(np.argmax(array, axis=0))
```

```
[4 4 4 4]
```

```
# numpy.sort()
```

```
import numpy as np

array = np.array([
    [3, 7, 1],
    [10, 3, 2],
    [5, 6, 7]
])
print(array)
print()
```

```
[[ 3  7  1]
 [10  3  2]
 [ 5  6  7]]
```

```
print(np.sort(array, axis=None))
```

```
[ 1  2  3  3  5  6  7  7 10]
```

```
print(np.sort(array, axis=1))
```

```
[[ 1  3  7]
 [ 2  3 10]
 [ 5  6  7]]
```

```
print(np.sort(array, axis=0))
```

```
[[ 3  3  1]
 [ 5  6  2]
 [10  7  7]]
```

```
#numpy.argsort()
```

```
import numpy as np

array = np.array([28, 13, 45, 12, 4, 8, 0])
print(array)

print(np.argsort(array))
```

```
[28 13 45 12  4  8  0]
[6 4 5 3 1 0 2]
```

```
# numpy.mean()
```

```
import numpy as np
```

```
list = [
    np.array([3, 2, 8, 9]),
    np.array([4, 12, 34, 25, 78]),
    np.array([23, 12, 67])
]

result = []
for i in range(len(list)):
    result.append(float(np.mean(list[i])))

print(result)
```

```
[5.5, 30.6, 34.0]
```

```
# add row
```

```
import numpy as np

array = np.array([
    [3, 2, 8],
    [4, 12, 34],
    [23, 12, 67]
])

newRow = np.array([2, 1, 8])
newArray = np.vstack((array, newRow))
print(newArray)
```

```
[[ 3  2  8]
 [ 4 12 34]
 [23 12 67]
 [ 2  1  8]]
```

```
import numpy as np

array = np.array([
    [3, 2, 8],
    [4, 12, 34],
    [23, 12, 67]
])

newColumn = np.array([2, 1, 8])
newArray = np.column_stack((array, newColumn))
print(newArray)
```

```
[[ 3  2  8  2]
 [ 4 12 34  1]
 [23 12 67  8]]
```

```
import numpy as np

array = np.array([3, 6, 7, 2, 5, 1, 8])
reversedArray = np.flipud(array)
print(reversedArray)
```

```
[8 1 5 2 7 6 3]
```

```
import numpy as np

matrix1 = [
    [3, 4, 2],
    [5, 1, 8],
    [3, 1, 9]
]

matrix2 = [
    [3, 7, 5],
    [2, 9, 8],
    [1, 5, 8]
]

result = np.dot(matrix1, matrix2)
print(result)
```

```
[[19 67 63]
 [25 84 97]
 [20 75 95]]
```

```
import numpy as np
```

```
n = 8

# Create a nxn matrix filled with 0
matrix = np.zeros((n, n), dtype=int)

# fill 1 with alternate rows and column
matrix[::2, 1::2] = 1
matrix[1::2, ::2] = 1

# Print the checkerboard pattern
for i in range(n):
    for j in range(n):
        print(matrix[i][j], end=" ")
    print()
```

```
0 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0
0 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0
0 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0
0 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0
```

Start coding or generate with AI.